The Central Distribution: A Minimax-KL Approach to Noninformative Priors and Feature Selection

Behrouz
aybehrouz@gmail.com

June 18, 2023

1 Introduction

1.1 The Optimal Strategy

We begin by revisiting a fundamental decision-making problem, and how it is treated within the framework of Bayesian decision theory.

Consider an environment where an agent must select an action from a set of available actions. The loss associated with each action depends on the unknown state of the environment and can be quantified by a real number. Specifically, let a denote an action, and x represent the unknown environment state. The loss function $\ell(a, x)$ quantifies the loss for each action.

Furthermore, we assume that an evidence e of this unknown environment state exists, and the agent can leverage this evidence to determine the best action. In other words, our agent aims to find an optimal strategy, where a strategy defines the distribution of each action conditioned on the observed evidence: $S(A \mid E)$.

According to Bayesian decision theory, the optimal strategy is the one that minimizes the expected loss:

$$S_{\text{opt}} = \arg\min_S \sum_a \sum_{x,e} \ell(a, x) S(a \mid e) P^*(x, e),$$

where P^* is the joint probability distribution of the environment state and the evidence.

It can be shown that there always exists a deterministic optimal strategy. That is, we can consider a strategy as a function of the evidence, rather than as a conditional distribution:

$$S_{\text{opt}} = \arg\min_S \sum_{x,e} \ell(S(e), x) P^*(x, e).$$

A strategy that, for each evidence e, selects the action minimizing the expected loss conditioned on e, is optimal:

$$\begin{aligned} S_{\text{opt}}(e) &= \arg\min_a \mathbb{E}_{P^*} \left[\ell(a, x) \mid e \right] \\ &= \arg\min_a \sum_x \ell(a, x) P^*(x \mid e) \cdot \end{aligned} \tag{1}$$

Now, suppose that we decide to use a different evidence E' instead of E. Could this change potentially improve our strategy? Within the framework of Bayesian decision theory, we can easily answer this question: To compare any two strategies, we can simply compare their expected loss. The strategy with the lower expected loss is the better one.

Note that, in reality, we cannot freely choose any random variable as our evidence. We can only choose evidence that is a deterministic function of the evidence provided by the environment, allowing us to determine its value after observing the original evidence. Utilizing a strategy that uses a simpler evidence, resembles feature selection in machine learning. For instance, in a learning task where our evidence is an image, we might substitute the image with the first three coefficients of its Fourier transform. Naturally, these coefficients are a function of the original evidence.

Interestingly, if we assume $E' = f(E)$, it turns out that, according to our Bayesian framework, the optimal strategy for E is always better than any strategy that uses E'.

From a certain perspective, it could be argued that adopting a strategy which uses simpler evidence may lead to ignoring some parts of useful available information, and therefore is not justified by Bayesian decision theory.

The key point here is that the space of strategies using E' is a subset of the strategies that use E. When we are searching for the best strategy based on E, we are implicitly evaluating all simpler forms as well. It is possible that this is the reason the Bayesian framework does not favor the simpler evidence E'.

Plausibly, if we choose an appropriate prior distribution, when we are finding the optimal strategy, the relevant information should be automatically extracted from the evidence by the Bayesian framework. In other words, an appropriate prior should enable feature selection. However, many commonly used noninformative priors, such as the flat prior, do not exhibit feature selection characteristics.

Throughout this discussion, we will explore a systematic approach for finding a noninformative prior that also exhibit feature selection characteristics.

1.2 Divergence

Suppose that for a given environment, we have specified a probability measure P. In particular, we are interested in the distribution it induces on a random variable X, denoted $P(X)$.

Now, imagine we ask an expert—a person familiar with the environment—to evaluate our choice of P. The expert may hold different beliefs and instead favor another probability measure Q. From the expert's perspective, evaluating P involves assessing the potential loss incurred by using the (in their view) incorrect distribution $P(X)$ when the true distribution is believed to be $Q(X)$.

Let us assume that a real number can be used to measure this loss, and denote it as $\mathbb{D}\left[Q(X) \,\|\, P(X)\right]$. It seems reasonable, as P gets closer to Q this measure decreases, and increases when P diverges from Q. Therefore, $\mathbb{D}$ effectively is a measure of the divergence of $P(X)$ from $Q(X)$, when Q is the believed true probability measure.

Interestingly, if we make certain assumptions about the properties of this divergence measure, it turns out that—up to a constant factor—it must be equal to the Kullback–Leibler (KL) divergence.

Alternatively, by adopting a more minimal set of properties, one can prove that accepting Shannon's entropy as a measure of the uncertainty implies that KL divergence can be considered as an appropriate method for measuring divergence between distributions.

Theorem 1. *Let X be a discrete random variable. Suppose $\mathbb{D}\left[Q(X) \,\|\, P(X)\right]$ is some divergence measure that quantifies the divergence of $P(X)$ from $Q(X)$. Consider the following properties:*

Consistency with entropy *Let Ω be a continuous random variable, and Q be a probability measure. If $U(\Omega)$ is the uniform distribution with the same support as $Q(\Omega)$, then we have*

$$\mathbb{D}\left[Q(\Omega) \,\|\, U(\Omega)\right] = \mathbb{H}\left[U(\Omega)\right] - \mathbb{H}\left[Q(\Omega)\right],$$

where $\mathbb{H}$ denotes Shannon's entropy.

Additivity *For any two discrete random variables X, Y and probability measures P, Q*

$$\mathbb{D}\left[Q(X,Y) \,\|\, P(X,Y)\right] = \mathbb{D}\left[Q(X) \,\|\, P(X)\right] + \sum_{x} \mathbb{D}\left[Q(Y|x) \,\|\, P(Y|x)\right] Q(x).$$

If $\mathbb{D}\left[\,\cdot\,\right]$ satisfies these properties, then it is the Kullback-Leibler divergence.

Proof. Let $P(X)$ and $Q(X)$ be two arbitrary distributions for a discrete random variable X. We define a new continuous random variable $\varOmega$ and let

$$p(\varOmega = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \leq \omega \leq P(X = x), \\ 0 & \text{otherwise}\cdot \end{cases}$$

Notice that both $P(\varOmega|X)$ and $P(\varOmega, X)$ are uniform distributions, and

$$\mathbb{H}\left[P(\varOmega|x)\right] = \ln P(x),\ \ \mathbb{H}\left[P(\varOmega, X)\right] = 0\cdot$$

We define $Q(\varOmega|x)$ to be an arbitrary distribution that has the same support as $P(\varOmega|x)$. This ensures that $Q(\varOmega, X)$ and $P(\varOmega, X)$ have the same support as well. Thus, we have

$$\begin{aligned} \mathbb{D}\left[Q(\varOmega, X) \parallel P(\varOmega, X)\right] &= \mathbb{H}\left[P(\varOmega, X)\right] - \mathbb{H}\left[Q(\varOmega, X)\right] \\ &= \sum_x \int q(\omega, x) \ln q(\omega, x) d\omega \\ &= \sum_x Q(x) \int q(\omega \mid x) \ln q(\omega, x) d\omega, \end{aligned}$$

and

$$\begin{aligned} \mathbb{D}\left[Q(\varOmega|x) \parallel P(\varOmega|x)\right] &= \mathbb{H}\left[P(\varOmega|x)\right] - \mathbb{H}\left[Q(\varOmega|x)\right] \\ &= \ln P(x) + \int q(\omega \mid x) \ln q(\omega \mid x) d\omega\cdot \end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$\begin{aligned} \mathbb{D}\left[Q(X) \parallel P(X)\right] &= \mathbb{D}\left[Q(\varOmega, X) \parallel P(\varOmega, X)\right] - \sum_x \mathbb{D}\left[Q(\varOmega|x) \parallel P(\varOmega|x)\right] Q(x) \\ &= \sum_x Q(x) \left(\int q(\omega \mid x) \ln \frac{q(\omega, x)}{q(\omega \mid x)} d\omega - \ln P(x) \right) \\ &= \sum_x Q(x) \left(\ln Q(x) \int q(\omega \mid x) d\omega - \ln P(x) \right) \\ &= \sum_x Q(x) \ln \frac{Q(x)}{P(x)}\cdot \end{aligned}$$

□

In this theorem, the first property simply states that the entropy of a distribution should be related to the divergence of the uniform distribution from it. The more the divergence is, the less the entropy must be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(X,Y)$ from $Q(X,Y)$ is a measure of the generic loss of using $P(X,Y)$ when we believe $Q(X,Y)$ is the true distribution. To calculate this loss, we can assume that X happens before Y. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with the probability $Q(x)$, it would also include the loss of using $P(Y \mid X = x)$. This property indicates that for combining these individual terms we should use addition.

We now briefly review some KL divergence properties needed in subsequent sections.

Lemma 1. *Let X be a random variable, and let P, Q, and R be probability distributions over X. The Kullback–Leibler divergence satisfies the following inequality:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel P(X)\right] \geq \sum_x Q(x) \ln \frac{R(x)}{P(x)}.$$

Proof. The Kullback–Leibler divergence is nonnegative and is defined when R is a probability distribution:

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel R(X)\right] \geq 0.$$

By adding $\sum_x Q(x) \ln \frac{R(x)}{P(x)}$ to both sides of the above inequality, the desired inequality follows. □

Theorem 2. *Let Q_1, Q_2 and P be probability distributions. If for some $0 \leq \alpha \leq 1$ we have $Q = \alpha Q_1 + (1-\alpha)Q_2$, then:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel P(X)\right] \leq \alpha \mathbb{D}_{\mathrm{KL}}\left[Q_1(X) \parallel P(X)\right] + (1-\alpha)\mathbb{D}_{\mathrm{KL}}\left[Q_2(X) \parallel P(X)\right].$$

That is, the KL divergence is convex in its first argument.

Proof. The result follows from Lemma 1 after expanding the left-hand side of the inequality using the definition of KL divergence. □

The KL divergence plays a key role in the notion of central distribution, which we introduce in the next section.

2 Central Distribution

Consider a random variable X. In probabilistic modeling, our knowledge about X is represented by a probability distribution, which serves as a basis for making inferences.

However, in many situations, our information may be too limited to justify choosing a single distribution. Instead, we may only be able to specify a set of plausible distributions. Such a set represents some form of *uncertainty* or *partial knowledge*: although it does not identify a single distribution, it constrains the range of plausible probabilistic models.

A well-known example of this idea is the exchangeability assumption used in de Finetti's theorem. A sequence of random variables $X_1, X_2, \ldots$ is said to be exchangeable, if the ordering of the variables carries no information. That is, the joint probability distribution remains invariant under permutations of the variables. This assumption imposes a constraint, effectively introducing a set of plausible distributions. Thus, exchangeability can be seen as a specific form of partial knowledge.

In this work, we adopt the view that a set of probability distributions can represent a meaningful state of partial knowledge, and that such set-valued uncertainty may itself be used as the basis for inference.

More generally, partial knowledge may possess a hierarchical structure. In the simplest case, uncertainty is represented by a single set of candidate distributions. In richer settings, however, this uncertainty may be organized into nested collections, where elements of a set may themselves be sets of distributions. Such nesting can continue recursively, yielding a hierarchical uncertainty structure.

Crucially, the choice of hierarchical organization is itself informative: grouping distributions into subsets expresses that certain alternatives are regarded as belonging to a common epistemic category and should therefore be aggregated before being compared with alternatives in other groups. Consequently, the hierarchy encodes information beyond the collection of leaf distributions alone.

This hierarchical organization may be interpreted as introducing multiple layers of latent structural decisions, where higher-level choices restrict the admissible lower-level alternatives.

Within a Bayesian framework, inference ultimately requires a single probability measure. Therefore, to use hierarchical partial knowledge in Bayesian inference, we require a principled method for transforming a nested set-valued uncertainty representation into a single probability distribution while preserving the informational structure encoded by the hierarchy.

We adopt the principle that the probability measure used to represent prior beliefs about the environment should depend only on the available information about the environment itself, and not on agent specific considerations such as the loss function or the informativeness of observations. We refer to this modeling assumption as *the principle of objectivity.*

From a minimax perspective, when uncertainty is represented by a set of candidate distributions, a natural representative distribution is one that minimizes the worst-case divergence to the members of the set. Such a distribution may be viewed as the most conservative or robust single-distribution summary of the available partial knowledge.

Definition 2.1 (Central Distribution)**.** Let X be a random variable and $\mathcal{Q}$ be a set of probability distributions over X. We say a distribution C is a *central distribution* of $\mathcal{Q}$ for X, if for every probability distribution P, we have

$$\sup_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, C(X)\right] \leq \sup_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right], \tag{2}$$

where $\mathbb{D}_{\mathrm{KL}}$ denotes the Kullback–Leibler (KL) divergence.

Note that, under this definition, the central distribution of $\mathcal{Q}$ is not necessarily an element of $\mathcal{Q}$. That is, the distribution that best represents the set in the specified sense may lie outside the set it summarizes.

Sometimes partial knowledge is represented by a hierarchical structure of sets of distributions. Our current definition of the central distribution does not account for this case. The following recursive definitions extend the concept to handle such situations. While not directly about feature selection, this extension plays a key role in enabling the connection between central distributions and feature selection, a connection that will become clearer later.

Definition 2.2 (Hierarchy of Distributions)**.** A *hierarchy of distributions* over X is defined recursively as follows:

- Any set of probability distributions over X is a hierarchy of distributions over X.
- Any set whose elements are hierarchies of distributions over X is itself a hierarchy of distributions over X.

Intuitively, a central distribution acts as a representative or summary of a set of distributions. While it may not belong to the set itself, it reflects the overall behavior of the set in a way that is meaningful for inference and also for evaluating a distribution based on our partial knowledge.

The concept of a central distribution is not intended to yield a unique object; rather, all distributions satisfying the definition can be regarded as equally good representatives. However, when multiple central distributions exist, subsequent constructions should not depend on an arbitrary choice among them. To ensure that the notion of central divergence is well-defined independently of which representative is selected, we define it using the best divergence attained over the set of all central distributions.

Definition 2.3 (Central Divergence)**.** For any distribution P, we define the *central divergence* of P from a hierarchy of distributions $\mathcal{H}$, with respect to a random variable X, as follows:

$$\mathbb{D}_{\mathrm{CD}}\left[\mathcal{H} \parallel P(X)\right] = \inf_{C} \mathbb{D}_{\mathrm{KL}}\left[C(X) \parallel P(X)\right],$$

where C denotes a central distribution of $\mathcal{H}$ for X.

The above definition applies only when the hierarchy is a single set of distributions. The following definition extends the construction recursively to general hierarchical uncertainty structures.

Definition 2.4. Let X be a random variable, and let $\mathcal{H}^*$ be a set whose elements are hierarchy of distributions over X. We say C^* is a central distribution of $\mathcal{H}^*$, if for every probability distribution P we have:

$$\sup_{\mathcal{H} \in \mathcal{H}^*} \mathbb{D}_{\mathrm{CD}}\left[\mathcal{H} \parallel C^*(X)\right] \leq \sup_{\mathcal{H} \in \mathcal{H}^*} \mathbb{D}_{\mathrm{CD}}\left[\mathcal{H} \parallel P(X)\right],$$

where $\mathbb{D}_{\mathrm{CD}}\left[\mathcal{H} \parallel \cdot\,\right]$ refers to the central divergence as defined in Definition 2.3.

A simple example illustrates that hierarchical organization can affect the resulting representative distribution even when the underlying admissible sets remain unchanged.

Let X be a random variable with three possible outcomes x_1, x_2, x_3. For each i, let S_i denote the set of all probability distributions over X that assign zero probability to x_i.

By symmetry, the central distribution of each set S_i is the uniform distribution over its support:

$$C_{S_1} = \left(0, \frac{1}{2}, \frac{1}{2}\right), \qquad C_{S_2} = \left(\frac{1}{2}, 0, \frac{1}{2}\right), \qquad C_{S_3} = \left(\frac{1}{2}, \frac{1}{2}, 0\right).$$

If the three sets S_1, S_2, S_3 are treated symmetrically as elements of a single set, then their central distribution is

$$C_{\{S_1, S_2, S_3\}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

Now consider instead a hierarchical organization in which S_1 and S_2 are grouped together, while S_3 forms its own separate branch.

Taking the central distribution of these two representatives yields

$$\begin{aligned} C_{\{\{S_1,S_2\},S_3\}} &= \frac{1}{2}C_{\{S_1,S_2\}} + \frac{1}{2}C_{S_3} \\ &= \frac{1}{4}C_{S_1} + \frac{1}{4}C_{S_2} + \frac{1}{2}C_{S_3} \\ &= \left(\frac{3}{8}, \frac{3}{8}, \frac{1}{4}\right). \end{aligned}$$

Thus, although the underlying admissible sets are identical in both constructions, the resulting representative distribution differs.

This demonstrates that hierarchical organization carries informational content beyond the collection of leaf sets alone. Consequently, flattening a hierarchy into a single unstructured set may discard meaningful epistemic structure.

In many learning problems, uncertainty concerns not only the parameters of a predictive model but also its structural form. In particular, when the relevance of potential features is unknown, each candidate feature subset determines a distinct family of distributions in which the target variable depends only on that feature subset.

Feature selection may therefore be viewed as uncertainty over a collection of distributions. The hierarchical central distribution framework provides a natural mechanism for representing and aggregating such uncertainty. By organizing candidate features hierarchically, one may encode preferences over structural hypotheses, such as favoring sparse subsets, grouping related attributes, or privileging particular interactions.

Candidate feature subsets typically induce overlapping families of admissible distributions. Distributions exhibiting simpler dependency structure may belong simultaneously to many such families, whereas more specific dependency patterns belong to fewer. As a result, hierarchical aggregation through central distributions can induce a preference toward structurally simpler models by assigning greater representation to distributions compatible with multiple structural hypotheses.

With the notion of hierarchical central distributions established, we now develop the theoretical machinery needed to characterize and compute them. The following results provide optimization-theoretic characterizations of central distributions that form the foundation of the subsequent analysis.

Definition 2.5 (Interior Distribution)**.** Let $\mathcal{Q}$ be a set of probability distributions over a random variable X. A distribution R is called an *interior* *distribution* of $\mathcal{Q}$ if, for every probability distribution P, there exists some $Q \in \mathcal{Q}$ such that

$$\mathbb{D}_{\mathrm{KL}}\left[R(X) \,\|\, P(X)\right] \leq \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right],$$

Theorem 3. *Any distribution that can be expressed as a convex combination of elements of $\mathcal{Q}$ is an interior distribution of $\mathcal{Q}$.*

Proof. Let $R = \sum_i \alpha_i Q_i$, where $Q_i \in \mathcal{Q}$, $\alpha_i \geq 0$, and $\sum_i \alpha_i = 1$.

Fix an arbitrary probability distribution P. By the convexity of KL divergence in its first argument, we have:

$$\mathbb{D}_{\mathrm{KL}}\left[R(X) \,\|\, P(X)\right] \leq \sum_i \alpha_i \mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \,\|\, P(X)\right] \cdot$$

Now, note that the right-hand side is a weighted average of the KL divergences. So, there must be at least one index j such that

$$\mathbb{D}_{\mathrm{KL}}\left[R(X) \,\|\, P(X)\right] \leq \mathbb{D}_{\mathrm{KL}}\left[Q_j(X) \,\|\, P(X)\right] \cdot$$

Since $Q_j \in \mathcal{Q}$, the condition in Definition 2.5 is satisfied, and R is indeed an interior distribution of $\mathcal{Q}$. □

Theorem 4. *Let R be an interior distribution of $\mathcal{Q}$. If C is a central distribution of $\mathcal{Q}$, it is also a central distribution of $\mathcal{Q} \cup \{R\}$ and vice versa.*

Proof. This theorem follows directly from the definition of an interior distribution, which implies that for any probability distribution P, we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] = \sup_{Q \in \mathcal{Q} \cup \{R\}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] \cdot \qquad \square$$

One important consequence of this theorem is that, in order to identify a central distribution within a parametric model, it suffices to restrict attention to the set of likelihood distributions induced by different values of the model parameters.

The next theorem is a standard result that is typically used in the derivation of central distributions.

Theorem 5. *Let X and T be two random variables, and let $\mathcal{Q}$ be a set of distributions over the pair (X, T). Assume that for every $Q \in \mathcal{Q}$, we have*

$$Q(X, T) = R(X \mid T) Q(T),$$

where R is a fixed conditional distribution. If C is a central distribution of $\mathcal{Q}$ for T, then $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X, T).

Proof. Let C be a central distribution of $\mathcal{Q}$ for T. For an arbitrary probability distribution P we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X,T) \,\|\, R(X \mid T)C(T) \right] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X,T) \,\|\, R(X \mid T)P(T) \right] \cdot \tag{3}$$

Since the KL divergence is nonnegative, for every probability distribution P and every $Q \in \mathcal{Q}$, we have

$$\mathbb{D}_{\mathrm{KL}} \left[Q(X,T) \,\|\, R(X \mid T)P(T) \right] \leq \mathbb{D}_{\mathrm{KL}} \left[Q(X,T) \,\|\, P(X,T) \right],$$

which implies

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X,T) \,\|\, R(X \mid T)P(T) \right] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X,T) \,\|\, P(X,T) \right] \cdot$$

Therefore, combining this inequality with Inequality 3, we conclude that $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X,T). $\square$

The central distribution is essentially the solution to a convex-concave game. Since the solution to a convex-concave game corresponds to a saddle point, by defining the appropriate objective function, we can characterize the central distribution using standard convex optimization techniques.

Definition 2.6 (Saddle Point)**.** Consider a function $f : \mathcal{W} \times \mathcal{Z} \to \mathbb{R}$. We refer to a pair (w^*, z^*) as a *saddle point* for f, if we have

$$f(w^*, z) \leq f(w^*, z^*) \leq f(w, z^*),$$

for all $w \in \mathcal{W}$ and $z \in \mathcal{Z}$.

Next, we will establish the connection between the saddle point and the central distribution. One significant insight provided by the next theorem is that it suggests inference and searching for a central distribution can be combined and performed simultaneously when finding a saddle point.

Theorem 6. *Let X be a discrete random variable, and $\mathcal{Q}$ be a finite set of distributions over X. Consider the following function:*

$$f(P, R) = \sum_i \mathbb{D}_{\mathrm{KL}} \left[Q_i(X) \,\|\, P(X) \right] R(i),$$

where $Q_i \in \mathcal{Q}$, and P, R are probability distributions.

If (P^*, R^*) is a saddle point of f, then P^* is a central distribution of $\mathcal{Q}$ for X.

Proof. Since X is a discrete random variable, $\mathcal{Q}$ has a central distribution for X. Let C be a central distribution of $\mathcal{Q}$ for X, and (P^*, R^*) be a saddle point of f. As C is a probability distribution, the point (C, R^*) lies within the domain of f. By the definition of a saddle point, this implies that

$$f(P^*, R^*) \leq f(C, R^*)\cdot \tag{4}$$

Because R^* is a probability distribution, the definition of f implies:

$$f(C, R^*) \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, C(X)\right]\cdot$$

On the other hand,

$$f(P^*, R^*) = \max_{R} f(P^*, R) = \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P^*(X)\right]\cdot$$

Thus, Inequality 4 implies

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P^*(X)\right] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, C(X)\right]\cdot$$

Since C is a central distribution, we must have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P^*(X)\right] = \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, C(X)\right],$$

and P^* is also a central distribution of $\mathcal{Q}$ for X. □

Lemma 2. *Let f be the function defined in theorem 6. The point (P^*, R^*) is a saddle point for f, if and only if there are constants λ, η and a function μ, such that for every i, x the following conditions hold:*

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} - \mu(i) = \lambda, \tag{5}$$

$$P^*(x) = \sum_i R^*(i) Q_i(x) \tag{6}$$

$$\sum_i R^*(i) = 1, \tag{7}$$

$$R^*(i) \geq 0, \tag{8}$$

$$\mu(i) \geq 0,$$

$$\mu(i) R^*(i) = 0\cdot \tag{9}$$

Furthermore, for a fixed probability distribution R, a distribution $\hat{P}$ minimizes $f(\,\cdot\,, R)$, if and only if for all x it satisfies

$$\hat{P}(x) = \sum_i Q_i(x) R(i)\cdot \tag{10}$$

Proof. Similar to f, we define $\hat{f}$ as follows:

$$\hat{f}(P,R) = \sum_i R(i) \sum_x Q_i(x) \ln \frac{Q_i(x)}{P(x)},$$

where the requirement that P and R be probability distributions is omitted. This enables us to formulate the saddle point of f as the solution to a constrained optimization problem defined for $\hat{f}$.

Let (P^*, R^*) be a simultaneous solution to the following two optimization problems:

$$\begin{array}{lll} \text{minimize} & \hat{f}(P,R) & \text{w.r.t. } P \\ \text{subject to} & \displaystyle\sum_x P(x) = 1, & \\ & & \\ \text{maximize} & \hat{f}(P,R) & \text{w.r.t. } R \\ \text{subject to} & R(i) \geq 0, & \text{for all } i \\ & \displaystyle\sum_i R(i) = 1 & \end{array} \tag{11}$$

Clearly, if (P^*, R^*) exists, it will be a saddle point for f.

Both of these optimization problems are convex [2] and differentiable. To establish this, consider that all the constraint functions are affine, and therefore are concave-convex and differentiable.

Additionally, $\hat{f}$ is differentiable function with respect to both R and P. For any fixed P, $\hat{f}(P, \cdot)$ is an affine function and is concave.

On the other hand, for a fixed solution to the second optimization problem denoted by $\hat{R}$, we can write

$$\hat{f}(P,\hat{R}) = c - \sum_{i,x} \hat{R}(i) Q_i(x) \ln P(x), \tag{12}$$

where c is not a function of P. Since $\hat{R}$ is a solution to the second optimization problem, it satisfies the corresponding constraints and is therefore a valid probability distribution. Hence, for all i, x we have $\hat{R}(i)Q_i(x) \geq 0$, and $\hat{f}(\cdot, \hat{R})$ is the negative of a linear combination of concave $\ln P(x)$ functions with nonnegative coefficients. This function is convex.

Thus, both optimization problems are convex and differentiable, with affine constraints. The constraints simply restrict the feasible set to the set of probability distributions, and it is always possible to find feasible points that meet all the constraints. Therefore, Slater's condition is satisfied, and the KKT conditions provide necessary and sufficient conditions for optimality [2]. In other words, If there exists a point (P^*, R^*) that satisfies the KKT conditions for both problems, it will be a saddle point for f, and vice versa.

The KKT conditions are as follows for every x, i and constants λ, η:

$$\sum_i R^*(i) Q_i(x) = \eta P^*(x) \tag{13}$$

$$\sum_x P^*(x) = 1, \tag{14}$$

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} - \mu(i) = \lambda,$$

$$\sum_i R^*(i) = 1, \tag{15}$$

$$R^*(i) \geq 0,$$

$$\mu(i) \geq 0,$$

$$\mu(i) R^*(i) = 0\cdot$$

By summing Equation 13 over x and using Equation 15, we have

$$\eta = 1\cdot$$

Thus, we get Equation 6 which also implicitly implies Equation 14.

To prove the second part of the lemma, we observe that Equations 13 and 14 are the KKT conditions for the minimization problem in 11, when R is a probability distribution. Therefore, these conditions provide necessary and sufficient conditions for a point to be a minimizer of $f(\,\cdot\,, R)$, where R is a fixed distribution. $\square$

Lemma 3. *Let $\mathcal{Q}$ be a finite set of probability distributions over a discrete random variable X. Define the function g as:*

$$g(R) = \sum_i \mathbb{D}_{\mathrm{KL}} \left[Q_i(X) \parallel R(X) \right] R(i),$$

where $Q_i \in \mathcal{Q}$, and R is a probability distribution such that

$$R(X, I) = Q_i(X) R(I)\cdot$$

Let f be the function defined in theorem 6. If (P^*, R^*) is a saddle point for f, then the joint distribution $Q_i(X) R^*(I)$ is a maximizer of g.

Proof. Let (P^*, R^*) be a saddle point of f. By Lemma 2, P^* satisfies Equation 6, and thus:

$$g(R^*) = f(P^*, R^*)\cdot$$

Now, let R be an arbitrary probability distribution. From Lemma 2, it follows that $g(R)$ indeed corresponds to the minimum of $f(\,\cdot\,, R)$, which implies

$$g(R) \leq f(P^*, R)\cdot$$

On the other hand, since (P^*, R^*) is a saddle point, we also have

$$f(P^*, R) \leq f(P^*, R^*)\cdot$$

Combining the two inequalities, we get:

$$g(R) \leq g(R^*)\cdot$$

Since R was an arbitrary distribution, this shows that $g(R^*)$ is the maximum of g, as desired. □

The following theorem gives a standard sufficient condition for centrality in the KL-minimax setting which is the rationale behind the term "central distribution." Exactly like the center of a circle, the central distribution is equidistant from the boundary.

Theorem 7 (Centrality)**.** *Let X be a discrete random variable, and let $\mathcal{Q}$ be a finite set of distributions over X. Suppose C is a convex combination of elements of $\mathcal{Q}$, with all weights strictly positive. If for some constant λ and every $Q \in \mathcal{Q}$ we have:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, C(X)\right] = \lambda,$$

then C is a central distribution of $\mathcal{Q}$ for X.

Proof. Lemma 2 provides a set of sufficient conditions for a saddle point. Note that more restrictive conditions can also be imposed to derive an alternative set of sufficient conditions. We can replace Inequality 8 with the more restrictive $R^*(i) > 0$ condition. By Equation 9, that implies $\mu(i) = 0$, and Lemma 2 conditions can be simplified to

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} = \lambda,$$
$$P^*(x) = \sum_i R^*(i) Q_i(x),$$

which must hold for every $Q_i \in \mathcal{Q}$ and some strictly positive probability distribution R^*.

Since C satisfies these condition, Lemma 2 and Theorem 6 imply that C is a central distribution of $\mathcal{Q}$ for X. $\square$

Lemma 4. *Assuming the conditions of Theorem 7 and using the same λ, if P is a convex combination of elements of $\mathcal{Q}$, then there exists some $Q_0 \in \mathcal{Q}$ such that*

$$\mathbb{D}_{\mathrm{KL}}\left[Q_0(X) \,\|\, P(X)\right] \leq \lambda\cdot$$

Proof. We proceed by contradiction. Suppose that for every $Q \in \mathcal{Q}$, we have $\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] > \lambda$. This implies that

$$g(P) > \lambda = g(C),$$

where C is the distribution specified in Theorem 7, and g is the function defined in Lemma 3. Note that $g(P)$ is defined, because P is a convex combination of elements of $\mathcal{Q}$.

However, by the conditions of Theorem 7, the distribution C corresponds to a saddle point of f. This leads to a contradiction, since Lemma 3 asserts that a saddle point of f must be a maximizer of g. Therefore, our assumption must be false. $\square$

These theorems were proved under the assumption that $\mathcal{Q}$ is finite; however, they can be extended to cases where $\mathcal{Q}$ can be indexed by a finite-dimensional real vector θ:

$$\mathcal{Q} = \left\{Q_\theta \mid \theta \in \Theta\right\}.$$

We can treat θ as a random variable and interpret Q_θ as a conditional distribution $Q(x \mid \theta)$. Then, the weight function, $R(i)$, will essentially become a prior distribution over θ.

The next theorem highlights a formal connection between the central-distribution framework and Bernardo's reference prior methodology [1].

Theorem 8. *Let X be a discrete random variable, and let $\mathcal{Q}$ be a set of probability distributions over X, indexed by a finite-dimensional random vector θ. Let c be a strictly positive prior distribution for θ such that for $Q(x \mid \theta) \in \mathcal{Q}$ we have*

$$c(\theta) \propto \exp\left\{\sum_x Q(x \mid \theta) \ln c(\theta \mid x)\right\}\cdot \tag{16}$$

Then, $C(x) = \int Q(x \mid \theta) c(\theta) d\theta$ is a central distribution of $\mathcal{Q}$ for X. We also refer to c as a central prior for X.

Proof. From Equation 16, it follows that there exists a constant λ such that

$$c(\theta) = \exp\left\{-\lambda + \sum_x Q(x \mid \theta) \ln \frac{Q(x \mid \theta)c(\theta)}{C(x)}\right\}.$$

Since $c(\theta)$ is strictly positive, we have

$$\begin{aligned} c(\theta) &= \exp\left\{-\lambda + \mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right] + \ln c(\theta)\right\} \\ &= c(\theta) \exp\left\{-\lambda + \mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right]\right\}, \end{aligned}$$

which implies

$$\mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right] = \lambda\cdot$$

Since this holds for all θ, and c is strictly positive, by Theorem 7 it follows that C is a central distribution of $\mathcal{Q}$ for X. □

The above condition defines an implicit fixed-point characterization of a central prior. This result should be interpreted as a sufficient characterization rather than a general existence or uniqueness theorem.

Suppose we want to model an exchangeable sequence of random variables. Let S_n denote the sequence of the first n variables: $S_n = (X_1, X_2, \ldots, X_n)$. As discussed earlier, exchangeability can be interpreted as a form of partial knowledge, which allows us to represent our uncertainty using a central distribution over S_n.

However, this central distribution generally depends on the sequence length n. According to the principle of objectivity, our beliefs should not depend on the specifics of our experimental setup, and in particular, should not vary with the length of the sequence.

It is important to note that a distribution defined over a longer sequence induces marginal distributions over its shorter subsequences. This observation suggests that, in order to satisfy the principle of objectivity, we may instead determine a central distribution over an infinite exchangeable sequence, from which the distributions of all finite initial segments can be derived.

This requires constructing a central distribution that is well-defined over S_∞.

Definition 2.7. Let $\{S_n\}_{n=1}^\infty$ be a sequence of random variables and $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each S_n. Assume C_n is a central distribution for S_n. Let C be a probability measure that defines a distribution over every S_n. C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^\infty$, if we have

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{KL}}\left[C_n(S_n) \,\|\, C(S_n)\right] = 0\cdot$$

Intuitively, this definition states that as $n \to \infty$ the distribution that C induces over S_n becomes increasingly similar to a central distribution for S_n. In other words, C is a central distribution for S_∞.

Our next goal is to extend Theorem 5 so that it also applies in this asymptotic setting, thereby providing a characterization of central distributions for infinite sequences.

Theorem 9. *Let $\{S_n\}_{n=1}^\infty$ and $\{T_n\}_{n=1}^\infty$ be two sequences of random variables. Let $\mathcal{Q}$ be a set of probability measures, where each $Q \in \mathcal{Q}$ defines a joint distribution over (S_n, T_n). Assume there exists a natural number M and a fixed conditional distribution R such that, for all $Q \in \mathcal{Q}$ and $n \geq M$,*

$$Q(S_n, T_n) = R(S_n \mid T_n)Q(T_n).$$

Let C be a central distribution of $\mathcal{Q}$ for $\{T_n\}_{n=1}^\infty$ (Definition 2.7). Define a probability measure C^* by

$$C^*(S_n, T_n) = R(S_n \mid T_n)C(T_n), \quad n \geq M.$$

Then C^* is a central distribution for $\{(S_n, T_n)\}_{n=1}^\infty$.

Proof. By Theorem 5, for each $n \geq M$ we have

$$C_n(S_n, T_n) = R(S_n \mid T_n)C_n(T_n),$$

where $C_n(T_n)$ is a central distribution for T_n, and $C_n(S_n, T_n)$ is a central distribution for (S_n, T_n). Therefore,

$$\begin{aligned}
\mathbb{D}_{\text{KL}}\left[C_n(S_n, T_n) \,\|\, C^*(S_n, T_n)\right] &= \mathbb{D}_{\text{KL}}\left[R(S_n \mid T_n)C_n(T_n) \,\|\, R(S_n \mid T_n)C(T_n)\right] \\
&= \mathbb{D}_{\text{KL}}\left[C_n(T_n) \,\|\, C(T_n)\right].
\end{aligned}$$

Since C is a central distribution for $\{T_n\}_{n=1}^\infty$ we have

$$\lim_{n\to\infty} \mathbb{D}_{\text{KL}}\left[C_n(T_n) \,\|\, C(T_n)\right] = 0.$$

It follows that

$$\lim_{n\to\infty} \mathbb{D}_{\text{KL}}\left[C_n(S_n, T_n) \,\|\, C^*(S_n, T_n)\right] = 0,$$

so C^* is a central distribution for $\{(S_n, T_n)\}_{n=1}^\infty$. □

We now present an asymptotic version of Theorem 7, applicable to infinite sequences of random variables.

Theorem 10. *Let $\{S_n\}_{n=1}^{\infty}$ be a sequence of random variables. Suppose C is a weighted average of elements of $\mathcal{Q}$ with strictly positive weights, and there exists a sequence of constants $\{\lambda_n\}_{n=1}^{\infty}$ such that*

$$\lim_{n\to\infty} \sup_{Q\in\mathcal{Q}} \left|\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right| = 0,$$

then C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$.

Proof. For the purpose of this proof, we additionally assume that for each S_n, there exists a central distribution C_n that satisfies the conditions of Theorem 7. That is, there exists a constant μ_n such that for every $Q \in \mathcal{Q}$

$$\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C_n(S_n)\right] = \mu_n.$$

For each n, by the definition of central distribution, we have

$$\sup_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] \geq \mu_n, \tag{17}$$

and, by Lemma 4, it also follows that

$$\inf_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] \leq \mu_n. \tag{18}$$

On the other hand, for each n we have

$$\left|\sup_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right| \leq \sup_{Q\in\mathcal{Q}} \left|\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right|,$$

and

$$\left|\inf_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right| \leq \sup_{Q\in\mathcal{Q}} \left|\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right|.$$

This implies that the left hand sides of Equations 17 and 18 both converge to λ_n. Now, by the squeeze theorem, the sequence $\{\mu_n - \lambda_n\}_{n=1}^{\infty}$ must converge to 0. Therefore, for any arbitrary $Q \in \mathcal{Q}$, we have

$$\begin{aligned}
\lim_{n\to\infty} \sum_{s_n} Q(s_n) \ln \frac{C_n(s_n)}{C(s_n)} &= \lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C_n(S_n)\right]\right) \\
&= \lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \mu_n + \lambda_n - \lambda_n\right) \\
&= \lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right) - \lim_{n\to\infty} (\mu_n - \lambda_n) \\
&= 0.
\end{aligned}$$

Since C_n satisfies the conditions of Theorem 7, $\mathbb{D}_{\mathrm{KL}}\left[C_n(S_n) \,\|\, C(S_n)\right]$ is a weighted average of $\sum_{s_n} Q(s_n) \ln \frac{C_n(s_n)}{C(s_n)}$, and we can conclude

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{KL}}\left[C_n(S_n) \,\|\, C(S_n)\right] = 0.$$

Therefore, C is a central distribution of $\mathcal{Q}$ for the sequence $\{S_n\}_{n=1}^{\infty}$. $\square$

Assume $\mathcal{Q}$ is a set of distributions that can be indexed by a parameter vector θ, and $\{T_n\}_{n=1}^{\infty}$ is a consistent estimator sequence for θ. Let $P(\theta \mid T_n = t)$ denote the posterior distribution obtained from an arbitrary smooth positive prior.

Under regular parametric conditions, posterior distributions based on consistent estimators become asymptotically insensitive to the choice of smooth positive prior.

Motivated by this phenomenon, we heuristically derive the asymptotic form of a central prior by considering the posterior density evaluated at the point $t = \theta$. A similar asymptotic construction also appears in [1].

Specifically, we define

$$c(\theta) \propto \lim_{n\to\infty} \frac{P(\theta \mid T_n = t)|_{t=\theta}}{\mu_n}, \tag{19}$$

where $\{\mu_n\}_{n=1}^{\infty}$ is a sequence of positive constants chosen so that the limit exists for every value of θ.

Let k denote the normalizing constant converting proportionality into equality. Since the logarithm is a continuous function, Equation 19 implies

$$\lim_{n\to\infty} \left(\ln \frac{P(\theta \mid T_n = t)|_{t=\theta}}{c(\theta)} - \ln \mu_n + \ln k \right) = 0,$$

Since T_n is a consistent estimator, the sampling distribution $Q\left(T_n \mid \theta\right)$ becomes increasingly concentrated around $T_n = \theta$ as $n \to \infty$. Furthermore, under the usual regularity conditions, the posterior distribution becomes asymptotically insensitive to the choice of smooth positive prior. Motivated by this asymptotic prior-insensitivity, we heuristically replace the original prior with $c(\theta)$, we obtain

$$\lim_{n\to\infty} \left(\sum_{t_n} Q(t_n \mid \theta) \ln \frac{Q(t_n \mid \theta)}{C(t_n)} - \ln \mu_n + \ln k \right) = 0\cdot$$

Letting

$$\lambda_n = \ln \mu_n - \ln k,$$

this suggests that, for each fixed θ:

$$\lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}}\left[Q\left(T_n \mid \theta\right) \| C(T_n)\right] - \lambda_n\right) = 0\cdot$$

Under additional regularity assumptions ensuring that this convergence is uniform over θ, the assumptions of Theorem 10 are satisfied. Thus, the above asymptotic argument suggests that $c(\theta)$ acts as a central prior of $\mathcal{Q}$ for the estimator sequence $\{T_n\}_{n=1}^{\infty}$.

A central distribution over an estimator cannot, by itself, be used to perform inference at the level of observed outcomes. To do so, we must convert it into a distribution over the primary variables of interest. Theorems 5 and 9 provide a simple method for this conversion.

Example 2.1 (Binomial Experiment)**.** Consider a binomial experiment with n Bernoulli trials. Denote the sequence of outcomes by S_n, and let k be the number of successes observed in S_n

Let $\mathcal{Q}$ be the set of distributions corresponding to different values of the success probability $\theta \in [0,1]$. We can consider θ as an index parameter for $\mathcal{Q}$, and we write $Q(S_n \mid \theta)$ to denote the distribution Q_θ applied to S_n. Explicitly,

$$Q(S_n \mid \theta) = \theta^k(1-\theta)^{n-k}\cdot$$

Define T_n to be the proportion of successes in n trials. Note that T_n is a consistent estimator for θ. Therefore, we can use Equation 19 to find a central distribution for the sequence $\{T_n\}_{n=1}^{\infty}$.

Assuming a uniform prior on θ, Bayes' rule gives the posterior distribution

$$P\left(\theta \,\middle|\, T_n = \tfrac{k}{n}\right) = (n+1)\binom{n}{k}\theta^k(1-\theta)^{n-k},$$

which is a Beta$(k+1, n-k+1)$ distribution. Using Stirling's formula, we derive its asymptotic form:

$$P\left(\theta \,\middle|\, T_n = \tfrac{k}{n}\right) \sim \frac{(n+1)n^{(n+\frac{1}{2})}}{\sqrt{2\pi}(n-k)^{(n-k+\frac{1}{2})}k^{(k+\frac{1}{2})}}\,\theta^k(1-\theta)^{n-k}\cdot$$

We now let $T_n \to \theta$:

$$P\left(\theta \mid T_n = \theta\right) \sim \frac{n+1}{\sqrt{2\pi n}}\,\theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}\cdot$$

Therefore, if we define

$$\mu_n = \frac{n+1}{\sqrt{2\pi n}},$$

the following limit exists:

$$\lim_{n\to\infty} \frac{P(\theta \mid T_n = \theta)}{\mu_n} = \theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}.$$

This implies that $c(\theta) \propto \theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}$ is a central prior for $\{T_n\}_{n=1}^{\infty}$.

Now we apply Theorem 9 to obtain a central distribution for $\{S_n\}_{n=1}^{\infty}$. Since T_n is a deterministic function of S_n, the joint distribution $Q(S_n, T_n)$ coincides with the distribution of S_n: it assigns probability $Q(S_n)$ when T_n is compatible with S_n, and zero otherwise. In other words, knowledge of (S_n, T_n) is equivalent to knowledge of S_n.

In the binomial experiment, all sequences with the same number of successes and failures have equal probability. Therefore,

$$Q(S_n \mid \theta) = Q\big(S_n,\, T_n = \tfrac{k}{n} \mid \theta\big) = \frac{1}{\binom{n}{k}} Q(T_n \mid \theta).$$

Let C^* denote the central distribution of $\{S_n\}_{n=1}^{\infty}$. By Theorem 9, the distribution induced by C^* on each S_n is given by:

$$C^*(S_n) = \frac{\Gamma\left(k+\frac{1}{2}\right)\Gamma\left(n-k+\frac{1}{2}\right)}{\Gamma(n+1)\,\Gamma\left(\frac{1}{2}\right)^2}.$$

When dealing with layered ignorance and attempting to determine the central distribution of a hierarchy of distributions, as defined in Definition 2.4, we typically encounter the case where the number of elements at a given level of the hierarchy is finite. In that setting, and under appropriate regularity conditions, the central distribution reduces to an equally weighted average of the corresponding representative central distributions.

Example 2.2 (Fused Multinomial Model)**.** Let $\{S_n\}_{n=1}^{\infty}$ be a sequence where each $S_n = (X_1, X_2, \ldots, X_n)$ represents n independent trials of an experiment with k possible outcomes, and $\mathcal{Q}$ be a set of probability distributions over S_n indexed by a real vector θ. Let $f : \mathcal{X} \to \mathcal{Y}$ be a deterministic function that maps each outcome of X_i to a label $Y_i = f(X_i)$, where $\mathcal{Y} = \{1, \ldots, m\}$. Assume that for all i and $Q_\theta \in \mathcal{Q}$, if $f(x_i) = f(x_i')$ we have

$$Q_\theta(x_1, x_2, \ldots, x_n) = Q_\theta(x_1', x_2', \ldots, x_n').$$

That is, the conditional distribution $Q_\theta(X_i \mid Y_i = y)$ is uniform over the preimage $f^{-1}(y)$.

We can consider $\theta = (\theta_1, \ldots, \theta_m)$ to be a probability vector over $\mathcal{Y}$, where each θ_y specifies the probability of observing label y. For a sequence S_n, let

n_y denote the number of times label $y \in \mathcal{Y}$ appears. Let $|f^{-1}(y)|$ denote the number of outcomes in $\mathcal{X}$ that are mapped to label y by f. Then the probability of observing S_n under $Q_\theta \in \mathcal{Q}$, denoted $Q(S_n \mid \theta)$, is:

$$Q(S_n \mid \theta) = \prod_{y=1}^{m} \left(\frac{1}{|f^{-1}(y)|} \cdot \theta_y \right)^{n_y}.$$

Now, consider the estimator sequence $T_n = \frac{1}{n}(n_1, n_2, \dots, n_m)$, which is a consistent estimator for θ. The conditional distribution $Q(S_n \mid T_n)$ is then given by:

$$Q(S_n \mid T_n) = \frac{1}{n!} \prod_{y=1}^{m} \frac{n_y!}{|f^{-1}(y)|^{n_y}}.$$

Note that this distribution is independent of Q_θ. Therefore, by Theorem 5, a central distribution for S_n can be obtained from a central distribution for T_n.

As $n \to \infty$, the distribution of T_n converges to a multivariate Gaussian distribution. Using this fact and Equation 19, we can determine a central prior for $\{T_n\}_{n=1}^{\infty}$:

$$c(\theta) \propto \prod_{y=1}^{m} \theta_y^{-\frac{1}{2}}$$

This is a Dirichlet prior and induces a Dirichlet-multinomial distribution for each T_n:

$$C(T_n) = \frac{\Gamma\left(\frac{m}{2}\right)\Gamma(n+1)}{\Gamma\left(n+\frac{m}{2}\right)} \prod_{y=1}^{m} \frac{\Gamma\left(n_y + \frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)\Gamma(n_y+1)}.$$

Note that here, C denotes a central distribution for T_∞, and $C(T_n)$ refers to the marginal distribution induced on T_n.

By Theorem 5, we can derive a central distribution for $\{S_n\}_{n=1}^{\infty}$, which induces the following distribution over each S_n:

$$C_f(S_n) = \frac{\Gamma\left(\frac{m}{2}\right)}{\Gamma\left(n+\frac{m}{2}\right)} \prod_{y=1}^{m} \frac{\Gamma\left(n_y + \frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)|f^{-1}(y)|^{n_y}}.$$

3 Illustrative Examples

We now present some examples of central distributions and demonstrate how they can be used for feature selection.

3.1 Attribute Selection

In this example, we consider a classification problem where the goal is to predict a target class, denoted by Y, from a set of observed nominal attributes $(A_1, A_2, \ldots, A_n)$. We assume that Y may depend on an unknown subset of the attributes, rather than the full set.

This reflects a form of sparsity in the relationship between the class and the attributes, where irrelevant features are ignored in the true generative model.

We can use the fused multinomial model of Example 2.2 to capture this relationship. As an example, assume that Y depends only on A_2, A_5. Then, we define the labeling function f as follows:

$$f(A_1, A_2, \ldots, A_n, Y) = (A_2, A_5, Y)\cdot$$

In a fused multinomial model with this labeling function, Y will be conditionally independent from other attributes, given A_2, A_5. Additionally, $P(A_1, A_2, \ldots, A_n \mid A_2, A_5)$ will be uniform.

For every possible subset of attributes a distinct fused multinomial model of this form is needed. Each model corresponds to a set of plausible distributions. Definition 2.4 provides a way to define a central distribution for situations like this one.

By Theorem 7 we know that this central distribution would be a weighted average of the central distributions of individual fused multinomial models. Since we have a finite number of models, these weights are uniform:

$$C_{\mathcal{Q}^*}(S_n) \propto \sum_i C_{f_i}(S_n)\cdot$$

To establish this, observe that we can construct a consistent estimator for the model indicator I by checking for the equality of different outcomes. Specifically, the estimator identifies outcomes with equal frequency in the data and returns the indicator for the model that supports this equality.

As shown in Equation 1, the optimal Bayesian decision rule depends on the chosen loss function. In this example, we adopt the zero-one loss, which leads to predicting the label with the highest conditional probability. These probabilities are computed using the central distribution we previously derived.

To see how this approach works in practice, we carried out a synthetic experiment with the following setup. We considered binary attributes and generated each attribute independently with equal probability for 0 and 1. The class label was defined as a deterministic function of a fixed subset of 3 out of 12 total attributes, as shown in Table 1. The same subset of relevant attributes and labeling function were used across all trials.

Table 1: Mappings of 3-bit inputs under functions f_1 and f_2

Input (bits)	f_1(Input)	f_2(Input)
000	a	A
001	b	B
010	c	A
011	d	C
100	e	D
101	f	D
110	g	D
111	h	E

For each experiment, we generated a training set of size 20 and a test set of size 60. A larger test set was used to reduce variability in performance estimates, taking advantage of the synthetic nature of the data. To assess robustness to irrelevant features, we progressively removed one attribute at a time from the full set of 12, starting with irrelevant ones and stopping just before removing any of the three relevant attributes.

At each step, we trained and tested two versions of the central distribution: one assuming a deterministic label function (CD-Det) and one allowing for a conditional (non-deterministic) label distribution (CD-Prob). Both versions are constructed by aggregating over 2^{12} fused multinomial models, corresponding to all subsets of the 12 attributes. As a baseline, we also trained a decision tree classifier (DTree). All methods were trained on the same training data and evaluated on the same test data. Each configuration was repeated for 100 independent trials, and we reported the average accuracy (correct predictions divided by total test instances) over these trials.

The results, shown in Figures 1 and 2, reveal notable differences in performance depending on the underlying class assignment function. For function f_1, both deterministic and probabilistic methods achieve high accuracy, with the deterministic version of the central distribution slightly outperforming the other. In contrast, function f_2 is more difficult to predict, leading to a clearer separation in performance between the two methods. This difficulty

Figure 1: Accuracy vs. Number of Attributes (Function f_1, 20 train samples)

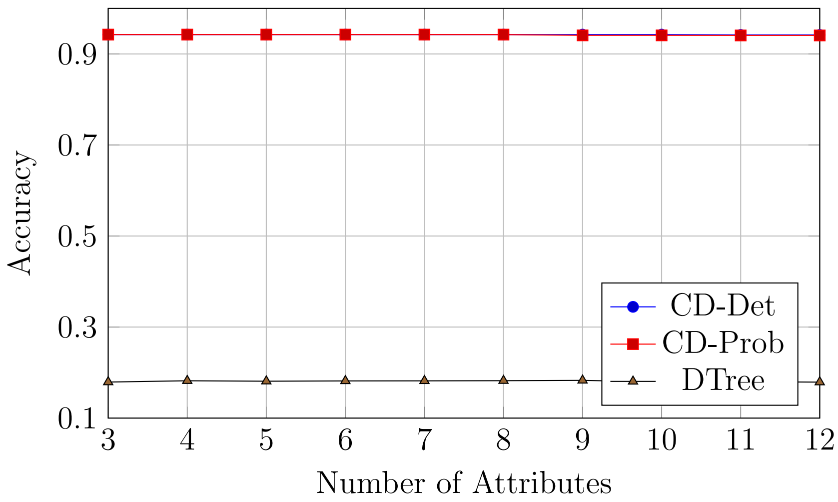
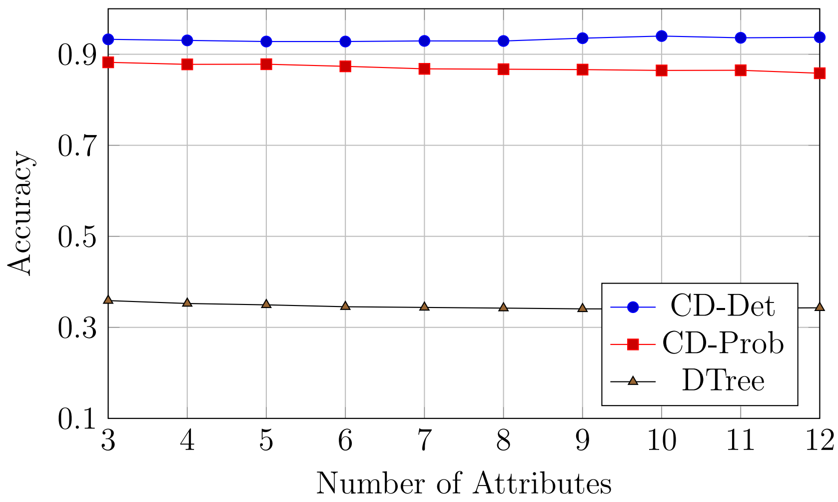
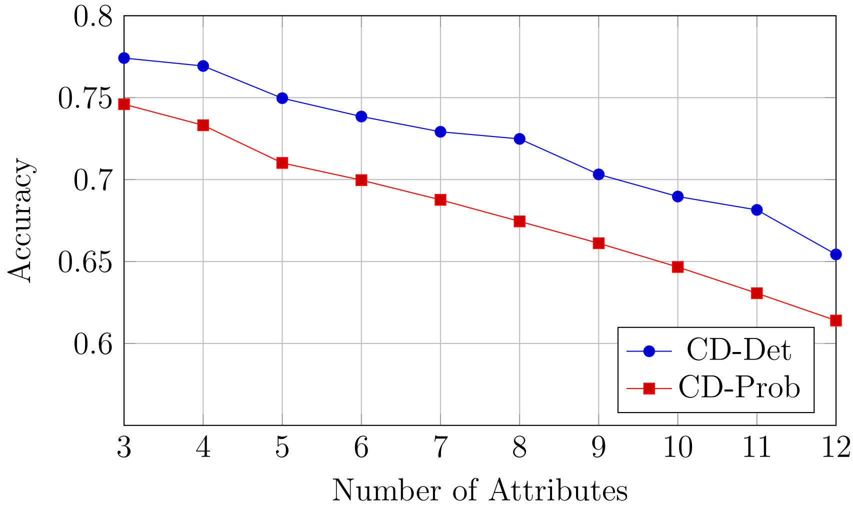


Figure 2: Accuracy vs. Number of Attributes (Function f_2, 20 train samples)



primarily arises from strong non-deterministic dependencies between subsets of input bits and the assigned label. For instance, when the first input bit of f_2 is 1, the label is often D, creating a misleading probabilistic association. In this case, the deterministic central distribution remains the most accurate overall, while the probabilistic version shows reduced performance due to misleading patterns.

Figure 3: Accuracy vs. Number of Attributes (Function f_2, 10 train samples)



While removing irrelevant attributes generally led to slight improvements in accuracy, the effect was not substantial in our primary experiments with 20 training samples. However, we observed that the benefit of attribute removal becomes more noticeable when the training set is smaller. In particular, additional experiments with 10 training samples showed a clearer performance gain as irrelevant features were removed, indicating that the impact of irrelevant attributes is more pronounced in low-data regimes (see Figure 3)

3.2 Latent Regression Model

We consider C classes (or datasets), indexed by $c = 1, \dots, C$. For each class c, we observe a data matrix $X^{(c)} \in \mathbb{R}^{N_c \times M}$, where each row $\boldsymbol{x}_i^{(c)} \in \mathbb{R}^M$ is associated with a D-dimensional latent variable $\boldsymbol{z}_i^{(c)} \in \mathbb{R}^D$.

Thus, each class is associated with a collection of latent variables

$$\boldsymbol{z}^{(c)} = \{\boldsymbol{z}_i^{(c)}\}_{i=1}^{N_c}, \qquad \boldsymbol{z}_i^{(c)} \in \mathbb{R}^D.$$

For each class c, the latent variables are assumed independent and identically distributed according to

$$\boldsymbol{z}_i^{(c)} \sim \mathcal{N}\big(\boldsymbol{\mu}^{(c)}, \tau_c^{-1} I_D\big)\,, \qquad c = 2, \ldots, C,$$

where $\boldsymbol{\mu}^{(c)} \in \mathbb{R}^D$ is the class-specific latent mean and $\tau_c > 0$ is a precision parameter controlling the latent dispersion.

To remove redundancy, we fix the latent distribution of the first class as a reference coordinate system. Specifically, we impose

$$\boldsymbol{\mu}^{(1)} = \mathbf{0}, \qquad \tau_1 = 1,$$

so that

$$\boldsymbol{z}_i^{(1)} \sim \mathcal{N}(\mathbf{0}, I_D).$$

This anchoring removes global translation and scaling symmetries without restricting the expressive capacity of the model. The latent geometry of each class is therefore interpreted relative to the reference class $c = 1$.

Let $\phi : \mathbb{R}^D \to \mathbb{R}^K$ denote a fixed nonlinear basis mapping. The design matrix of class c is defined as $Z^{(c)} \in \mathbb{R}^{N_c \times K}$ with entries

$$Z_{ik}^{(c)} = \phi_k(\boldsymbol{z}_i^{(c)}).$$

We collect the decoder weights in a shared matrix

$$W = [\boldsymbol{w}_1 \ \ldots \ \boldsymbol{w}_M] \in \mathbb{R}^{K \times M},$$

where each column $\boldsymbol{w}_j \in \mathbb{R}^K$ corresponds to the weights for the j-th observed dimension across all classes.

Conditioned on $\boldsymbol{z}^{(c)}$ and W, observations are generated as

$$p(X^{(c)} \mid \boldsymbol{z}^{(c)}, W) = \prod_{i=1}^{N_c} \prod_{j=1}^{M} \mathcal{N}\left(x_{ij}^{(c)} \;\middle|\; \phi(\boldsymbol{z}_i^{(c)})^\top \boldsymbol{w}_j, \beta_j^{-1}\right),$$

or equivalently,

$$\boldsymbol{x}_j^{(c)} \sim \mathcal{N}\big(Z^{(c)} \boldsymbol{w}_j, \beta_j^{-1} I_{N_c}\big)\,,$$

where $\boldsymbol{x}_j^{(c)}$ denotes the j-th column of $X^{(c)}$ and $\beta_j > 0$ is the observation precision of that column.

To regularize the decoder, we assume noninformative priors for the latent parameters $\{\boldsymbol{\mu}^{(c)}, \tau_c\}_{c=2}^{C}$, and place independent Gaussian priors on the columns of W:

$$p(W) = \prod_{j=1}^{M} \mathcal{N}\left(\boldsymbol{w}_j \mid \mathbf{0}, \alpha_j^{-1} I_K\right).$$

The full joint distribution factorizes as

$$p(W)\prod_{c=1}^{C} p(X^{(c)} \mid \boldsymbol{z}^{(c)}, W)p(\boldsymbol{z}^{(c)} \mid \boldsymbol{\mu}^{(c)}, \tau_c)p(\boldsymbol{\mu}^{(c)}, \tau_c).$$

This defines a shared nonlinear generative model in which all classes are coupled through the common decoder W, while retaining class-specific latent geometry via $(\boldsymbol{\mu}^{(c)}, \tau_c)$.

3.2.1 Variational Inference for the Single-Class Case

To simplify the exposition, we now consider the single-class case. Since only one class is present, we remove the superscript (c) from the notation for clarity.

Furthermore, we restrict the latent variable to be scalar,

$$\boldsymbol{z} = \{z_i\}_{i=1}^{N} \qquad z_i \in \mathbb{R},$$

so that

$$p(\boldsymbol{z} \mid \mu, \tau) = \prod_{i=1}^{N} \mathcal{N}(z_i \mid \mu, \tau^{-1}).$$

We treat μ and τ as unknown parameters to be inferred, using the standard joint Jeffreys prior for the mean and precision of a Gaussian model,

$$p(\mu, \tau) \propto \tau^{-1}.$$

The nonlinear basis mapping reduces to

$$\phi : \mathbb{R} \to \mathbb{R}^K,$$

and the design matrix $Z \in \mathbb{R}^{N\times K}$ is defined by

$$Z_{ik} = \phi_k(z_i).$$

We see the Bayesian network of this model in Figure 4. Exact posterior inference in this model is intractable due to the nonlinear dependence of the likelihood on $\boldsymbol{z}$. We therefore proceed by introducing a variational approximation.

In order to formulate a variational treatment of this model, we consider a variational distribution which factorizes between the latent variables and the parameters so that

$$q(\boldsymbol{z}, W, \mu, \tau) = q(\boldsymbol{z})q(W, \mu, \tau).$$

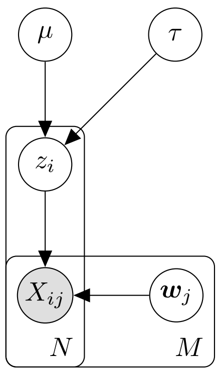


Figure 4: One class scalar latent regression model

First, let us consider the derivation of the update equation for the factor $q(W, \mu, \tau)$. The log of the optimized factor is given by

$$\ln q^*(W, \mu, \tau) = \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(W, \mu, \tau \mid \boldsymbol{z}, X)\right] + \text{const}.$$

As can be seen from Figure 4, the d-separation criterion implies that

$$\mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(W, \mu, \tau \mid \boldsymbol{z}, X)\right] = \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\mu, \tau \mid \boldsymbol{z})\right] + \sum_j \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\boldsymbol{w}_j \mid \boldsymbol{z}, X)\right].$$

Therefore, we have the following induced factorization for $q(W, \mu, \tau)$:

$$q(W, \mu, \tau) = q(\mu, \tau) \prod_{j=1}^{M} q(\boldsymbol{w}_j).$$

We obtain $q(\boldsymbol{w}_j)$ using the fact that each $\boldsymbol{w}_j$ is a weight vector of a Bayesian linear regression model:

$$\begin{aligned}\ln p(\boldsymbol{w}_j \mid \boldsymbol{z}, X) &= -\frac{\beta_j}{2}\left\|Z\boldsymbol{w}_j - \boldsymbol{x}_j\right\|^2 - \frac{\alpha_j}{2}\boldsymbol{w}_j^\top \boldsymbol{w}_j + \text{const} \\ &= -\frac{1}{2}\boldsymbol{w}_j^\top(\beta_j Z^\top Z + \alpha_j I)\boldsymbol{w}_j + \beta_j \boldsymbol{x}_j^\top Z\boldsymbol{w}_j + \text{const}.\end{aligned}$$

Taking the expectation with respect to $q(\boldsymbol{z})$, we obtain

$$\ln q^*(\boldsymbol{w}_j) = -\frac{1}{2}\boldsymbol{w}_j^\top(\beta_j \mathbb{E}\left[Z^\top Z\right] + \alpha_j I)\boldsymbol{w}_j + \beta_j \boldsymbol{w}_j^\top \mathbb{E}\left[Z\right]^\top \boldsymbol{x}_j + \text{const},$$

where expectations of matrices are taken element-wise. The resulting expression is quadratic in $\boldsymbol{w}_j$. Completing the square yields

$$q^*(\boldsymbol{w}_j) = \mathcal{N}(\boldsymbol{w}_j \mid \boldsymbol{\mu}_j, \Sigma_j).$$

The corresponding moments are

$$\mathbb{E}\left[\boldsymbol{w}_j\right] = \boldsymbol{\mu}_j,$$
$$\mathbb{E}\left[\boldsymbol{w}_j\boldsymbol{w}_j^\top\right] = \boldsymbol{\mu}_j\boldsymbol{\mu}_j^\top + \Sigma_j,$$

with

$$\Sigma_j^{-1} = \beta_j\mathbb{E}\left[Z^\top Z\right] + \alpha_j I,$$
$$\boldsymbol{\mu}_j = \beta_j\Sigma_j\mathbb{E}\left[Z\right]^\top \boldsymbol{x}_j.$$

Since the design matrix satisfies $Z_{ik} = \phi_k(z_i)$, we have

$$\mathbb{E}\left[Z^\top Z\right] = \sum_{i=1}^{N}\mathbb{E}\left[\phi(z_i)\phi(z_i)^\top\right].$$

Collecting the posterior means into a matrix gives

$$\mathbb{E}\left[W\right] = [\boldsymbol{\mu}_1 \ \ldots \ \boldsymbol{\mu}_M].$$

Moreover, to simplify later expressions, define the diagonal precision matrix

$$B = \operatorname{diag}(\beta_1, \ldots, \beta_M).$$

Then, we have

$$\sum_{j=1}^{M}\beta_j\mathbb{E}\left[\boldsymbol{w}_j\boldsymbol{w}_j^\top\right] = \mathbb{E}\left[W\right]B\mathbb{E}\left[W\right]^\top + \sum_{j=1}^{M}\beta_j\Sigma_j.$$

This quantity will appear in the update for the latent variables. Note that if $\alpha_j = 0$ for all j, the mean simplifies to

$$\mathbb{E}\left[W\right] = \mathbb{E}\left[Z^\top Z\right]^{-1}\mathbb{E}\left[Z\right]^\top X.$$

We now consider the update equation for the factor $q(\mu, \tau)$:

$$\begin{aligned}\ln q^*(\mu,\tau) &= \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\mu,\tau \mid \boldsymbol{z})\right] + \text{const}\\ &= -\frac{\tau}{2}\left(\sum_i\mathbb{E}\left[z_i^2\right] - 2\mu\sum_i\mathbb{E}\left[z_i\right] + N\mu^2\right)\\ &\quad + \left(\frac{N}{2} - 1\right)\ln\tau + \text{const}.\end{aligned}$$

We define

$$\overline{z} = \frac{1}{N}\sum_i \mathbb{E}\left[z_i\right],$$
$$\overline{zz} = \frac{1}{N}\sum_i \mathbb{E}\left[z_i^2\right].$$

We observe that $q^*(\mu, \tau)$, for $N > 1$ follows a Normal–Inverse–Gamma distribution:

$$q^*(\mu, \tau) = \mathcal{N}\left(\mu \;\middle|\; \overline{z}, \frac{1}{N\tau}\right) \mathrm{Gamma}\left(\tau \;\middle|\; \frac{N-1}{2}, \frac{N}{2}(\overline{zz} - \overline{z}^2)\right).$$

This implies

$$\mathbb{E}\left[\tau\right] = \frac{(N-1)}{N(\overline{zz} - \overline{z}^2)},$$
$$\mathbb{E}\left[\mu\tau\right] = \overline{z}\mathbb{E}\left[\tau\right].$$

Finally, we derive the update equation for $q(\boldsymbol{z})$:

$$\begin{aligned}
\ln q^*(\boldsymbol{z}) &= \mathbb{E}_{q(W,\mu,\tau)}\left[\ln p(X \mid \boldsymbol{z}, W) + \ln p(\boldsymbol{z} \mid \mu, \tau)\right] + \text{const} \\
&= \mathbb{E}_{q(W)}\left[\ln p(X \mid \boldsymbol{z}, W)\right] + \mathbb{E}_{q(\mu,\tau)}\left[\ln p(\boldsymbol{z} \mid \mu, \tau)\right] + \text{const.}
\end{aligned}$$

We expand the terms $\ln p(X \mid \boldsymbol{z}, W)$ and $\ln p(\boldsymbol{z} \mid \mu, \tau)$. All expressions that are independent of $\boldsymbol{z}$ are treated as constants.

$$\begin{aligned}
\ln p(X \mid \boldsymbol{z}, W) &= \sum_j \ln \mathcal{N}(\boldsymbol{x}_j \mid Z\boldsymbol{w}_j, \beta_j^{-1} I) \\
&= -\frac{1}{2}\sum_j \beta_j \left\|Z\boldsymbol{w}_j - \boldsymbol{x}_j\right\|^2 + \text{const} \\
&= \sum_i \left(-\frac{1}{2}\phi(z_i)^\top \sum_j \beta_j \boldsymbol{w}_j \boldsymbol{w}_j^\top \phi(z_i) + \phi(z_i)^\top \sum_j \beta_j X_{ij} \boldsymbol{w}_j\right) + \text{const.}
\end{aligned}$$

$$\begin{aligned}
\ln p(\boldsymbol{z} \mid \mu, \tau) &= -\frac{\tau}{2}\sum_i (z_i - \mu)^2 + \text{const} \\
&= \sum_i \left(-\frac{\tau}{2} z_i^2 + \tau\mu z_i\right) + \text{const.}
\end{aligned}$$

Substituting these expansions into the variational update expression and taking the expectation yields:

$$\ln q^*(\boldsymbol{z}) = \sum_i \ln q^*(z_i),$$

where

$$\ln q^*(z_i) = \left(-\frac{1}{2}\phi(z_i)^\top \sum_j \beta_j \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top\right] \phi(z_i) + \phi(z_i)^\top \mathbb{E}\left[W\right] B X_{i,:}^\top\right)$$
$$+ \left(-\frac{1}{2}\mathbb{E}\left[\tau\right] z_i^2 + \mathbb{E}\left[\tau\mu\right] z_i\right) + \text{const.}$$

Here $X_{i,:}$ denotes the i-th row of the observation matrix X.

Because the update expression separates as a sum over i, the optimal factor $q(\boldsymbol{z})$ factorizes as $\prod_i q(z_i)$. Define

$$S^{-1} = \sum_j \beta_j \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top\right],$$
$$\boldsymbol{m}_i = S\mathbb{E}\left[W\right] B X_{i,:}^\top .$$

Then, up to normalization,

$$q^*(z_i) \propto \mathcal{N}\left(\phi(z_i) \mid \boldsymbol{m}_i, S\right) \mathcal{N}\left(z_i \;\middle|\; \mathbb{E}\left[\mu\right], \mathbb{E}\left[\tau\right]^{-1}\right).$$

Note that since the basis mapping is nonlinear, this does not generally imply that $q^*(z_i)$ is Gaussian in z_i.

As seen previously, the update equations require expectations of the form

$$\mathbb{E}\left[\phi_i(z_k)\phi_j(z_k)\right], \quad \mathbb{E}\left[\phi_i(z_k)\right], \quad \mathbb{E}\left[z_k\right], \quad \mathbb{E}\left[z_k^2\right],$$

for all relevant indices i, j, k. These expectations typically do not admit closed forms and must be approximated numerically.

References

[1] José M. Bernardo. Reference analysis. In D.K. Dey and C.R. Rao, editors, *Bayesian Thinking*, volume 25 of *Handbook of Statistics*, pages 17–90. Elsevier, 2005.

[2] Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.